United States Patent	12416538
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Zhou; Zhen et al.

System and method for testing a specimen

Abstract

A system and a method of testing a specimen. The system includes an endcap, a plurality of bars, a plurality of strain gauges, and a gas gun. The endcap includes a first surface and a second surface opposite to the first surface. The first surface is curved. Each bar is disposed in contact with the second surface of the endcap and extends along a longitudinal axis. Each strain gauge is disposed on a surface of a corresponding bar from the plurality of bars. At least one strain gauge is disposed on the surface of each bar. The gas gun is configured to fire a specimen towards the first surface of the endcap such that the specimen impacts the first surface at an oblique angle relative to the longitudinal axis.

Inventors: **Zhou; Zhen (Derby, GB), Macdougall; Duncan A. (Derby, GB), Reed; Julian M. (Derby, GB)**

Applicant: **ROLLS-ROYCE plc** (London, GB)

Family ID: **1000008821463**

Assignee: **Rolls-Royce PLC (London, GB)**

Appl. No.: **18/183999**

Filed: **March 15, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20230314249 A1	Oct. 05, 2023

Foreign Application Priority Data

GB	2204670	Mar. 31, 2022

Publication Classification

Int. Cl.: **G01L5/1627** (20200101); **G01L1/22** (20060101)

U.S. Cl.:

CPC **G01L5/1627** (20200101); **G01L1/2218** (20130101); **G01L1/2287** (20130101);

Field of Classification Search

CPC: G01L (5/1627); G01L (1/2218); G01L (1/2287); G01L (5/14); G01M (5/0016); G01M (7/08)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5325701	12/1993	Zilliacus	N/A	N/A
6116077	12/1999	Albertini	73/12.05	G01N 3/30
7360393	12/2007	Abke	73/12.14	G01M 7/08
7845207	12/2009	Read	73/12.11	G01N 3/313
2018/0017065	12/2017	Hall et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
111551458	12/2019	CN	N/A
0859950	12/1996	EP	N/A
3043141	12/2015	EP	N/A
20150132682	12/2014	KR	N/A

OTHER PUBLICATIONS

Mou, H., Xie, J., Pei, H., Feng, Z., Geng, H.URL: doi.org/10.1016/J.AST.2020.106344XP86377453A, Ballistic impact tests and stacked shell simulation analysis of aramid fabric containment system, Aerospace Science and Technology 107 (2020) 106344, Nov. 11, 2020. cited by applicant
European search report dated Aug. 18, 2023, issued in EP Patent Application No. 23160860.5. cited by applicant
Great Britain Search Report dated Sep. 21, 2022, issued in GB Patent Application No. 2204670.0. cited by applicant

---

Primary Examiner: Kirkland, III; Freddie

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) This specification is based upon and claims the benefit of priority from United Kingdom patent application number GB 2204670.0 filed on Mar. 31, 2022, the entire contents of which is incorporated herein by reference.
BACKGROUND
Technical Field

(2) The present disclosure is related to a system and a method for testing a specimen.

Description of the Related Art

(3) Gas turbine engines typically include multiple rows of blades. Each row of blades is surrounded by a casing. There is a possibility of a blade being released and suffering an impact with the casing. Such an impact event is generally referred to as a blade release or containment event.

(4) To simulate such impact events using numerical techniques, such as Finite Element method, an accurate definition of a material of the blade is required. Typical practice is to generate material test data using small specimens over a range of stress states (e.g., tension, compression and shear), strain rates and temperatures relevant to the application.

(5) The material model, which is built using the specimen data, may be then validated using a component test. In case of blade containment scenarios, this is generally done using a blade crush test. In a conventional test setup, a missile impacts a tip of a blade, and loads are transmitted through a root of the blade into a Hopkinson bar tube, through an endcap. The conventional test setup may result in symmetric loading of the blade.

(6) However, the conventional test setup has various limitations. For example, the blade deformation may not be the same as a blade containment event. Further, the endcap may be of too low stiffness to transmit accurate loads into the Hopkinson bar. For long slender blades, the deformation/buckling characteristics caused by the conventional symmetric loading method may be overly sensitive to initial conditions leading to chaotic results. Therefore, in case of the conventional test setup, the load signature and blade crush behaviour are generally quite different from the containment event.

SUMMARY

(7) In one aspect, a system for testing a specimen is provided. The system includes an endcap, a plurality of bars, a plurality of strain gauges, and a gas gun. The endcap includes a first surface and a second surface opposite to the first surface. The first surface is curved. Each bar is disposed in contact with the second surface of the endcap and extends along a longitudinal axis. Each strain gauge is disposed on a surface of a corresponding bar from the plurality of bars. At least one strain gauge is disposed on the surface of each bar. The gas gun is configured to fire a specimen towards the first surface of the endcap such that the specimen impacts the first surface at an oblique angle relative to the longitudinal axis.

(8) Interaction with the endcap may cause the specimen (e.g., a compressor blade) to be loaded asymmetrically. The system may be able to accurately simulate a blade release or containment event. Specifically, the system may apply loading which is similar to a blade release or containment event. The asymmetric loading may produce results which are less sensitive to initial conditions (i.e., less chaotic).

(9) In some embodiments, the first surface of the endcap is concave. In some embodiments, the second surface of the endcap is planar.

(10) In some embodiments, the gas gun is further configured to fire the specimen towards the first surface of the endcap such that the specimen tangentially impacts the first surface. Firing the specimen tangentially towards the endcap may further improve the accuracy of the loading of the specimen.

(11) In some embodiments, each bar is attached to the endcap.

(12) In some embodiments, the system further includes a plurality of supporting members. Each supporting member supports a corresponding bar from the plurality of bars. At least one supporting member supports each bar along its length. In some embodiments, each supporting member is a support cable disposed at least partly around a circumference of the corresponding bar. A length of each support cable may be adjustable to ensure a correct orientation of each bar during testing.

(13) In some other embodiments, each supporting member may include a cylindrical bushing for supporting the corresponding bar.

(14) In some embodiments, the system further includes a bump-stop device spaced apart from an

end of each bar opposite to the endcap. The bump-stop device may prevent the bars from being subject to a gross displacement under loading to avoid unnecessary damage to the strain gauges and the bars.
(15) In some embodiments, the system further includes at least one camera for measuring one or more parameters. The one or more parameters includes at least one of: a speed of the specimen prior to impact with the first surface of the endcap; an orientation of the specimen prior to impact with the first surface of the endcap; a location of impact of the specimen; and a fracture of the specimen during impact. The one or more parameters may be used for validating a material model of the specimen.
(16) In some embodiments, the material of the endcap includes at least one of titanium, aluminium and steel.
(17) In some embodiments, the material of each bar includes at least one of titanium, aluminium and steel.
(18) In another aspect, a method for testing a specimen is provided. The method includes providing an endcap having a first surface and a second surface opposite to the first surface. The first surface is curved. The method further includes providing a plurality of bars disposed in contact with the second surface of the endcap and extending along a longitudinal axis. The method further includes providing at least one strain gauge on a surface of each bar. The method further includes firing a specimen towards the first surface of the endcap such that the specimen impacts the first surface at an oblique angle relative to the longitudinal axis. The method further includes measuring one or more loads transmitted to each bar by the endcap using the at least one strain gauge.
(19) Unlike conventional methods which generally causes symmetric loading, the method of the present disclosure may cause the specimen (e.g., a compressor blade) to be loaded asymmetrically due to the interaction with the endcap. The method may be able to accurately simulate a blade release or containment event. Specifically, the method may result in loading which is similar to a blade release or containment event. The asymmetric loading may produce results which are less sensitive to initial conditions (i.e., less chaotic).
(20) In some embodiments, the specimen is fired towards the first surface of the endcap such that the specimen tangentially impacts the first surface.
(21) In some embodiments, the method further includes measuring one or more parameters using at least one camera. The one or more parameters include at least one of: a speed of the specimen prior to impact with the first surface of the endcap, an orientation of the specimen prior to impact with the first surface of the endcap, a location of impact of the specimen; and a fracture of blade during impact.
(22) In some embodiments, the method further includes supporting each bar along its length by at least one supporting member.
(23) The skilled person will appreciate that except where mutually exclusive, a feature or parameter described in relation to any one of the above aspects may be applied to any other aspect. Furthermore, except where mutually exclusive, any feature or parameter described herein may be applied to any aspect and/or combined with any other feature or parameter described herein.

Description

DESCRIPTION OF THE DRAWINGS
(1) Embodiments will now be described by way of example only, with reference to the Figures, in which:
(2) FIG. **1** is a schematic perspective view of an exemplary system for testing a specimen;
(3) FIGS. **2**A-**2**D are different views of an endcap of the system of FIG. **1**;
(4) FIGS. **3**A-**3**B are different views of a bar of the system of FIG. **1**;

(5) FIGS. **4**A-**4**B are different views of a specimen before impact with an endcap;
(6) FIGS. **5**A-**5**B are different views of the specimen during impact with the endcap; and
(7) FIG. **6** is a flowchart of an exemplary method for testing a specimen.

DETAILED DESCRIPTION

(8) With reference to FIG. **1**, a system **100** for testing a specimen is provided, in accordance with an embodiment of the present disclosure. The system **100** includes an endcap **102**, a plurality of bars **104**, a plurality of strain gauges **106** and a gas gun **108**. The specimen can be a blade of a gas turbine engine. For example, the specimen can be a compressor blade, a turbine blade, a fan blade, a part of a blade, or a foreign body.

(9) The endcap **102** includes a first surface **202** and a second surface **204** (shown in FIG. **2**A) opposite to the first surface **202**. The first surface **202** is curved. Each bar **104** is disposed in contact with the second surface **204** of the endcap **102** and extends along a longitudinal axis "LA". In some embodiments, each bar **104** is attached to the endcap **102**. Each bar **104** may be attached to the endcap **102** by any suitable method, for example, fasteners, adhesives, welding, brazing, mechanical joints, or any combination thereof. In some embodiments, the bars **104** may be fused with the endcap **102**.

(10) Each strain gauge **106** is disposed on a surface **302** of a corresponding bar **104** from the plurality of bars **104**. At least one strain gauge **106** is disposed on the surface **302** of each bar **104**. In the illustrated embodiment of FIG. **1**, two strain gauges **106** are disposed on the surface **302** of each bar **104**. Each strain gauge **106** is used to measure strain at a specific location of the corresponding bar **104**. In some embodiments, each strain gauge **106** may be a resistive strain gauge including an insulating flexible backing which supports a metallic foil pattern. Each strain gauge **106** may be attached to the corresponding bar **104** by any suitable method, for example, an adhesive such as cyanoacrylate. Terminals of each strain gauge **106** may be electrically connected to a Wheatstone bridge (not shown) to measure strain. Different parameters of each strain gauge **106**, such as shape, material, dimensions, attachment location, attachment method etc., may vary based on application requirements.

(11) The gas gun **108** is configured to fire a specimen (not shown in FIG. **1**) towards the first surface **202** of the endcap **102** such that the specimen impacts the first surface **202** at an oblique angle relative to the longitudinal axis "LA". In the illustrated embodiment, the gas gun **108** is inclined obliquely relative to the longitudinal axis "LA". The gas gun **108** defines a gun axis "GA" along its length. The gun axis "GA" is inclined obliquely relative to the longitudinal axis "LA". An angle "A**1**" between the gun axis "GA" and the longitudinal axis "LA" may be substantially equal to the oblique angle at which the specimen impacts the first surface **202** of the endcap **102**. In some embodiments, the angle "A**1**" can be from about 110 degrees to about 130 degrees. In some embodiments, the gas gun **108** is further configured to fire the specimen towards the first surface **202** of the endcap **102** such that the specimen tangentially impacts the first surface **202**.

(12) As shown in FIG. **1**, the gas gun **108** has a substantially hollow tubular shape. However, the present disclosure is not limited by any shape of the gas gun **108**. In some embodiments, the gas gun **108** is a projectile-firing gun which is pneumatically powered by compressed air or any other gas. The air or gas can be pressurized mechanically.

(13) The system **100** further includes a plurality of supporting members **110**. Each supporting member **110** supports a corresponding bar **104** from the plurality of bars **104**. At least one supporting member **110** supports each bar **104** along its length. In the illustrated embodiment of FIG. **1**, each supporting member **110** is a support cable **112** disposed at least partly around a circumference of the corresponding bar **104**. In some embodiments, each support cable **112** is a braided twisted steel cable. Each support cable **112** may form a U-shaped configuration in order to support the corresponding bar **104**. In the illustrated embodiment, two support cables **112** are provided for each bar **104**. However, a number of support cables **112** for supporting each bar **104** may vary based on various factors, such as dimensions of each bar **104**, expected loads during

testing, etc. Each supporting member **110** further includes end supports **114** disposed at free ends of each support cable **112**. Each supporting member **110** may therefore include two end supports **114**. Each support cable **112** may be connected to a structure (not shown) by the corresponding end supports **114**. A cable length of each support cable **112** can be adjustable to ensure a correct orientation of each bar **104** during a test. In some other embodiments, the bars **104** may be alternatively or additionally supported using other methods, such as plain bearings or roller bearings.

(14) The system **100** further includes a bump-stop device **116** spaced apart from an end **304** of each bar **104** opposite to the endcap **102**. The end **304** may be a free end of each bar **104**. The end **304** of each bar **104** may not be constrained. The bump-stop device **116** may be retained or secured at a distance from the end **304** of each bar **104**. The bump-stop device **116** may prevent the bars **104** from being subject to a gross displacement under loading to avoid unnecessary damage to the bars **104** and the strain gauges **106**. A clearance "CL" between the bump-stop device **116** and the end **304** of each bar **104** may be varied as per application requirements. The bump-stop device **116** is embodied as a cuboidal member in FIG. **1**. However, a shape and dimensions of the bump-stop device **116** may vary based on the application. The bump-stop device **116** may be made of any suitable material, such as metal or metal alloy, plastic, elastomer or composite. The bump-stop device **116** may be an optional component, and testing may be carried out without the bump-stop device **116**.

(15) The system **100** further includes at least one camera **118** for measuring one or more parameters. In the illustrated embodiment of FIG. **1**, the system **100** includes two cameras **118** disposed at different orientations and locations. However, a number, locations and orientations of the cameras **118** can be chosen based on the testing and measurement requirements. Each camera **118** may be any optical device that can capture still images and/or moving images. Each camera **118** may be a digital camera or a film camera. In some embodiments, each camera **118** may be a high-speed camera for measuring one or more parameters related to the impact of the specimen with the endcap **102**. In some embodiments, the one or more parameters include at least one of: speed of the specimen prior to impact with the first surface **202** of the endcap **102**; an orientation of the specimen prior to impact with the first surface **202** of the endcap **102**; a location of impact of the specimen; and a fracture the specimen during impact.

(16) In some cases, the speed of the specimen prior to impact can be about 450 metres/second (m/s). Further, a crush load transmitted from the specimen to the endcap **102** due to impact can be less than or equal to about 300 kilonewtons (KN). In some embodiments, testing may be conducted in a vacuum chamber. The vacuum chamber may include portholes for viewing and/or recording the tests.

(17) FIGS. **2**A-**2**D illustrate various views of the endcap **102**. Referring to FIGS. **2**A-**2**D, the endcap **102** includes the first surface **202**, the second surface **204** opposite to the first surface **202**, a top surface **206**, a bottom surface **208** opposite to the top surface **206** and a pair of side surfaces **210**. The first and second surfaces **202**, **204** may be longitudinal surfaces of the endcap **102**. The first and second surfaces **202**, **204** may extend substantially perpendicular to the longitudinal axis "LA" (shown in FIG. **1**) of the bars **104**. The first surface **202** is curved. The first surface **202** is further bounded by a pair of longitudinal edges **202**A and a pair of lateral edges **202**B. A length of each longitudinal edge **202**A is greater than a length of each lateral edge **202**B. Further, the first surface **202** of the endcap **102** is concave. In some embodiments, the first surface **202** of the endcap **102** may have a partial cylindrical shape. In some embodiments, a radius of curvature "R**1**" of the first surface **202** is from about 500 millimetres (mm) to about 700 mm. In some embodiments, the radius of curvature "R**1**" is about 650 mm. In some embodiments, the second surface **204** is planar. Further, each of the top surface **206**, the bottom surface **208** and the side surfaces **210** is also planar.

(18) The endcap **102** further has a length "L", a width "W" and a depth "D". The length "L" may be from about 250 mm to about 400 mm. The width "W" may be from about 100 mm to about 200

mm. The depth “D” may be from about 50 mm to about 100 mm. In an example, the length “L” may be about 350 mm, the width “W” may be about 180 mm, and the depth “D” may be about 80 mm. A size of the endcap **102** may be changed based on various factors, such as duration of a crush event during testing and the size of the specimen.

(19) The endcap **102** further defines a plurality of holes **212**. In some embodiments, each hole **212** may be a blind hole. Each hole **212** may be drilled or bored into the endcap **102**. The holes **212** may be used to receive mechanical fasteners. One or more holes **212** may also be partially threaded. The first surface **202** may include four holes **212**. Two holes **212** may be proximal to the top surface **206**, while the other two holes **212** may be proximal to the bottom surface **208**. The second surface **204** may include two holes **212**. One of the holes **212** may be proximal to the top surface **206**, while the other hole **212** may be proximal to the bottom surface **208**. Each of the top and bottom surfaces **206**, **208** may include three holes **212**. Some of the holes **212** may communicate with each other. Each hole **212** may be substantially circular. A diameter of each hole **212** may be from about 2 mm to about 5 mm. In an example, the diameter of each hole **212** may be about 3 mm.

(20) The endcap **102** may be made of any suitable material, such as metal or metal alloy, plastic, elastomer or composite. In some embodiments, the endcap **102** is made of at least one of titanium, aluminium and steel.

(21) The endcap **102** can transmit loads from the impact to the bars **104** without any failure, substantial vibrations or loss of contact with the bars **104**. The endcap **102** may be designed to be stiff enough so that modal frequencies of the endcap **102** are above an impact frequency. The endcap **102** may be lightly supported to ensure contact with the bars **104**.

(22) FIGS. **3**A-**3**B illustrate various views of the bar **104**. The bar **104** may be a Hopkinson bar. Further, the bar **104** may be a solid cylindrical bar. A length “LB” of the bar **104** may be from about 3000 mm to about 4000 mm. A diameter “DB” of the bar **104** may be from about 50 mm to about 100 mm. In an example, the length “LB” of the bar **104** is about 3500 mm and the diameter “DB” of the bar **104** may be about 85 mm. A size of the bar **104** may be changed based on various factors, such as duration of a crush event during testing and the size of the specimen.

(23) The bar **104** may include one or more chamfered or rounded regions for contact with the endcap **102** and/or attachment with the strain gauges **106** (shown in FIG. **1**). The bar **104** may be made of any suitable material, such as metal or metal alloy, plastic, elastomer or composite. In some embodiments, the bar **104** is made of at least one of titanium, aluminium and steel.

(24) An exemplary crush test will be described with reference to FIGS. **4**A, **4**B, **5**A and **5**B using the system **100** described above.

(25) FIGS. **4**A and **4**B illustrate a specimen **602** prior to an impact with the endcap **102**. As shown in FIGS. **4**A and **4**B, the specimen **602** is a blade of a gas turbine blade. In an example, the specimen **602** is a compressor blade. The specimen **602** is fired from the gas gun **108** towards the first surface **202** of the endcap **102**. The specimen **602** may be about to impact one of the lateral edges **202**B of the endcap **102**. A tip of the specimen **602** may be located proximal to the lateral edge **202**B of the endcap **102**.

(26) FIGS. **5**A and **5**B illustrate the specimen **602** during impact with the first surface **202** of the endcap **102**. The tip of the specimen **602** may impact the first surface **202**. During impact, the specimen **602** may undergo deformation.

(27) The load signature of the crush test can be measured using the strain gauges **106** (shown in FIG. **1**) disposed on the bars **104**.

(28) The system **100** can be used to simulate a blade crush event behaviour when a blade containment event or release occurs. In case of compressor blades, the system **100** can accurately simulate compressor blade crush behaviour when a compressor containment event happens.

(29) The oblique orientation of the gas gun **108** relative to the longitudinal axis “LA” and the curved shape of the first surface **202** of the endcap **102** may ensure that the specimen **602** (i.e.,

blade) can arrive at the endcap **102** obliquely. In some cases, the specimen **602** can arrive tangentially at the endcap **102**. Interaction with the endcap **102** may cause the specimen **602** to be loaded asymmetrically. Loads are transmitted through the endcap **102** into the bars **104**. The strain gauges **106** on the surfaces **302** of the bars **104** can be used to measure the load signature of the crush event. High-speed photography using the cameras **118** can be used to measure the deformation history of the specimen **602**.
(30) The system **100** can be used to measure blade collapse load in an accurate manner. Further, the system **100** may crush a compressor blade in a manner similar to a compressor blade release event. Moreover, the system **100** may allow measurement of the dynamic deflections of the blade, including any locations and time of fracture. The bars **104** may also allow accurate measurement of the crush load. The load signature and the blade crush behaviour obtained using the system **100** may be close to a blade containment release event.
(31) In some embodiments, a projective velocity range (i.e., velocity of the specimen **602**) may be between 200 m/s and 500 m/s. In some embodiments, a projective mass range (i.e., mass of the specimen **602**) may be between 0.5 kg and 2 kg. In some embodiments, the number of bars **104** may be from 1 to 5. In some embodiments, the angle “A**1**” between the gun **108** and the bars **104** may be from 100 degrees to 150 degrees.
(32) FIG. **6** illustrates a method **900** for testing a specimen. The method **900** will be described with reference to the system **100** described above.
(33) At step **902**, the method **900** includes providing the endcap **102** having the first surface **202** and the second surface **204** opposite to the first surface **202**. The first surface **202** is curved.
(34) At step **904**, the method **900** further includes providing the plurality of bars **104** disposed in contact with the second surface **204** of the endcap **102** and extending along the longitudinal axis “LA”. In some embodiments, the method **900** further includes supporting each bar **104** along its length by at least one supporting member **110**.
(35) At step **906**, the method **900** further includes providing at least one strain gauge **106** on the surface **302** of each bar **104**.
(36) At step **908**, the method **900** further includes firing a specimen towards the first surface **202** of the endcap **102** such that the specimen impacts the first surface **202** at an oblique angle relative to the longitudinal axis “LA”. In some embodiments, the specimen is fired towards the first surface **202** of the endcap **102** such that the specimen tangentially impacts the first surface **202**.
(37) At step **910**, the method **900** further includes measuring one or more loads transmitted to each bar **104** by the endcap **102** using the at least one strain gauge **106**.
(38) In some embodiments, the method **900** further includes measuring one or more parameters using at least one camera **118**. In some embodiments, the one or more parameters include at least one of: speed of the specimen prior to impact with the first surface **202** of the endcap **102**; an orientation of the specimen prior to impact with the first surface **202** of the endcap **102**; a location of impact of the specimen; and a fracture the specimen during impact.
(39) It will be understood that the invention is not limited to the embodiments described above and various modifications and improvements can be made without departing from the concepts described herein. Except where mutually exclusive, any of the features may be employed separately or in combination with any other features and the disclosure extends to and includes all combinations and sub-combinations of one or more features described herein.

Claims

1. A system for testing a specimen, the system comprising: an endcap having a first surface and a second surface opposite to the first surface, wherein the first surface is curved; a plurality of bars, wherein each bar is disposed in contact with the second surface of the endcap and extends along a longitudinal axis; a plurality of strain gauges, wherein each strain gauge is disposed on a surface of

a corresponding bar from the plurality of bars, wherein at least one strain gauge is disposed on the surface of each bar; and a gas gun configured to fire a specimen towards the first surface of the endcap such that the specimen impacts the first surface at an oblique angle relative to the longitudinal axis.
2. The system of claim 1, wherein the first surface of the endcap is concave.
3. The system of claim 1, wherein the second surface of the endcap is planar.
4. The system of claim 1, wherein the gas gun is further configured to fire the specimen towards the first surface of the endcap such that the specimen tangentially impacts the first surface.
5. The system of claim 1, wherein each bar is attached to the endcap.
6. The system of claim 1, further comprising a plurality of supporting members, wherein each supporting member supports a corresponding bar from the plurality of bars, wherein at least one supporting member supports each bar along its length.
7. The system of claim 6, wherein each supporting member is a support cable disposed at least partly around a circumference of the corresponding bar.
8. The system of claim 1, further comprising a bump-stop device spaced apart from an end of each bar opposite to the endcap.
9. The system of claim 1, further comprising at least one camera for measuring one or more parameters, the one or more parameters comprising at least one of: a speed of the specimen prior to impact with the first surface of the endcap; an orientation of the specimen prior to impact with the first surface of the endcap; a location of impact of the specimen; and a fracture of the specimen during impact.
10. The system of claim 1, wherein a material of the endcap comprises at least one of titanium, aluminium and steel.
11. The system of claim 1, wherein a material of each bar comprises at least one of titanium, aluminium and steel.
12. A method for testing a specimen, the method comprising the steps of: providing an endcap having a first surface and a second surface opposite to the first surface, wherein the first surface is curved; providing a plurality of bars disposed in contact with the second surface of the endcap and extending along a longitudinal axis; providing at least one strain gauge on a surface of each bar; firing a specimen towards the first surface of the endcap such that the specimen impacts the first surface at an oblique angle relative to the longitudinal axis; and measuring one or more loads transmitted to each bar by the endcap using the at least one strain gauge.
13. The method of claim 12, wherein the specimen is fired towards the first surface of the endcap such that the specimen tangentially impacts the first surface.
14. The method of claim 12, further comprising measuring one or more parameters using at least one camera, the one or more parameters comprising at least one of: a speed of the specimen prior to impact with the first surface of the endcap; an orientation of the specimen prior to impact with the first surface of the endcap; a location of impact of the specimen; and a fracture of the specimen during impact.
15. The method of claim 12, further comprising supporting each bar along its length by at least one supporting member.